File Compression and Archives

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Accessibility Notice

This document is also available in HTML format at:

Link: `https://aholdengouveia.name/IntroLinux/labs/compressionarchive.html`

The HTML version provides enhanced accessibility features including keyboard navigation, screen reader support, responsive design, dark mode support, and high contrast options.

Learning Outcomes

By the end of this lab, you will be able to:

- Explain the difference between an archive and a compressed file
- Create, list, and extract tar archives
- Compress and decompress files using gzip
- Compare file sizes before and after compression

Lab Instructions

In this lab you will bundle files into an archive, compress them, and compare the results. You can do this on any Linux system.

How to set up for this lab

- Create a directory for this lab, and put at least 5 files in it.
- Put some text in each file, so there is something to compress.
- Include at least one larger file. For example, copy /etc/services into your lab directory.

Problems to solve

1. In your own words, what is the difference between an archive (such as a tar file) and a compressed file (such as a gzip file)?
2. Archives
   (a) Use tar to create an archive of your lab directory. Take a screenshot.
   (b) List the contents of your archive without extracting it. Take a screenshot.
   (c) Make a new directory and extract your archive into it. Use ls to show the files are there. Take a screenshot.
3. Compression
   (a) Use ls -l to check the size of your larger file.
   (b) Compress it with gzip, then check the size again. How much smaller is it? Take a screenshot.
   (c) Decompress it with gunzip. Take a screenshot.
4. Archive and compress in one step
   (a) Use tar to create a compressed archive (.tar.gz) of your lab directory. Take a screenshot.
   (b) Use ls -l to compare the sizes of your .tar and .tar.gz files. Which is smaller, and by how much? Take a screenshot.
   (c) Extract your .tar.gz into a new directory. Take a screenshot.
5. Give one example of when you would use an archive, one example of when you would use compression, and one example of when you would use both.

Deliverables

One file with the lab title, your name, and the date at the top.

- Each answer should be numbered to match the question.
- Make sure to list each command and the result.

- Make sure to take a screenshot of each successful command, labeled with which problem it solved. Your screenshots must show your prompt with your username, or a note with your name and the term.

Resources

References, a video, a PowerPoint and some notes are available at my website Link: `https://www.aholdengouveia.name/IntroLinux/compressionarchive.html`

My FAQ Linux playlist on YouTube, with demos of many of the commands and scripts in this class Link: `https://www.youtube.com/playlist?list=PLQOmuLC45eoQlIMH7oCNL6yFP7kWQubBo`

My video on how to use tar Link: `https://www.youtube.com/watch?v=x0EK6PmKRCU`